

Multimodal Retrieval Evaluation (Synthetic Sample)

This document is a synthetic sample used to demonstrate a dual-encoder multimodal retrieval system. All figures below are illustrative and do not describe any real deployment.

System design. The system uses two separate embedding towers. A text tower (a sentence embedding model) encodes text passages, and a CLIP image tower encodes images. The two towers produce vectors in different spaces, so they are stored in two separate indexes. A text query is embedded by both towers, which lets a short query retrieve both relevant passages and relevant images.

Text preparation. Document text is split into chunks of 500 characters with an overlap of 100 characters before it is embedded. The overlap keeps a claim that straddles a chunk boundary retrievable from either side.

Headline result. On the held-out set, the dual-encoder retrieval model reports an overall accuracy of 0.86. This is the combined configuration that uses both the text tower and the image tower together.

Results by configuration.

The chart on this page compares retrieval accuracy across three configurations: text-only, image-only, and the combined multimodal configuration. The combined configuration is the highest of the three, which is the motivation for using both towers rather than either one alone.

Reading the chart. Each bar is one configuration. Taller is better. The combined bar is tallest, the text-only bar is in the middle, and the image-only bar is the shortest for this synthetic data.

Retrieval accuracy by configuration

